

Grid Synchronization Phase-Locked Loop Strategy for Unbalance and Harmonic Distortion Conditions

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Abstract The synthesis, design and analysis of a three-phase phase-locked loop (PLL) algorithm under grid voltage uncertainties is presented. Unlike other techniques, the proposed strategy is simple but yet, robust against unbalanced and distorted voltage conditions. The method does not rely on the symmetry of the three-phase voltages, like conventional PLL implementation techniques. Under phase lock, the reference voltage and the output voltage are in synchronism, hence they cross the zero axis at the same time. The proposed PLL implementation exploits this fact, by counting the difference between the zero crossing times of the grid voltage and the output voltage waveform, and generating an error signal proportional to this difference. This error is fed into a PI regulator which forces it to zero over time. The proposed algorithm is compared against two popular PLL techniques. Simulation and experimental results show better performance of the proposed PLL implementation in unbalanced grid voltage conditions.

Keywords Phase-locked loop · Grid voltage distortion · Power electronics converters · Phase detection · Grid synchronization

1 Introduction

Renewable energy has, in the last decade, emerged as a viable tool for combating climate change, hence its widespread popularity in industry and academia. It is estimated that a third of global electricity generation will be supplied by renewable energy by 2035 (State Grid Corporation of China 2012; Hammons et al. 2000). A comprehensive survey of renewable energy systems, as they pertain to sustainable energy, was undertaken by Nehrir et al. (2011). New trends in power electronic technology are examined by Carrasco et al. (2006) for interfacing these renewable energy sources with the grid to maintain power supply reliability. The wind turbine is one of the many forms of renewable energy generator that has experienced rapid growth. However, interconnecting such a renewable energy system to the power grid requires a robust phase-locked loop (PLL) algorithm capable of dealing with uncertainties. In light of this, alternative synchronization strategies are necessary to fully exploit this energy source and keep pace with ever-growing energy needs.

A PLL is a closed-loop control system that modulates the phase angle of an output signal, so that it closely matches the phase angle of a reference input signal. It essentially minimizes the phase error between both signals, thus matching their oscillatory frequencies. The general technique of doing this is generating a voltage signal corresponding to the phase difference/frequency difference between the output signal and the reference signal. Thus, if the input signal is frequency- or phase-modulated, the output becomes the demodulated output of the PLL (Phipps et al. 2006).

A PLL implementation can be linear (S-domain), digital, or software. The general structure of any PLL consists of a phase detector (PD), a loop filter (LF), and a voltage controlled oscillator (VCO). The phase detector compares the phase angle of the reference signal, usually a

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